

A comparison of the effect of carbon dioxide and relative humidity sensitive ventilation systems on indoor air quality, hygrothermal and energy performance in an office room made of hemp concrete

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SUMMARY

One of the most important goals for building design is to satisfy the people occupying the building. One important part of this satisfaction is related to hygrothermal comfort, indoor air quality, for which the ventilation is very important factor to be taken care. A too low ventilation rate results in poor indoor air quality (IAQ) for the occupants and moisture risk, however, too high ventilation rate results in unnecessary energy consumption.

In this study, an IAQ-hygrothermal building model, in which the carbon dioxide is considered as indicator for evaluating IAQ, that takes into account the moisture buffering capacity of building envelope, indoor heat, moisture and carbon dioxide sources is presented. This model is implemented in the environment SPARK (Simulation Problem Analysis and Research Kernel) which is suitable to complex problems using finite difference technique with an implicit scheme. The model is then used to analyze the building performance in terms of hygrothermal comfort, energy consumption as well as IAQ of a room using carbon dioxide and relative humidity sensitive ventilation systems.

This study confirmed that the use of demand control ventilation system is a very efficient way to reduce the energy consumption. Compared to the CO₂ sensitive ventilation system the combined relative humidity sensitive (RHS) ventilation system and moisture buffering capacity of building envelope results in lower heating energy consumption but higher percent dissatisfied in terms of body odor and thermal comfort.

KEYWORDS: *Hygrothermal performance; Indoor Air Quality; Energy performance; Carbon dioxide sensitive ventilation system; Relative humidity sensitive ventilation system; SPARK*

1 INTRODUCTION

One of the most important goals for building design is to satisfy the people occupying the building. This satisfaction is related to the thermal comfort, indoor air quality, acoustical and visual comfort. Among the factors influencing perceived indoor air quality and human comfort, indoor relative humidity (RH) and CO₂ concentration are important parameters. High moisture and indoor CO₂ concentration levels can damage construction and have an adverse effect on health. High humidity harms materials, especially in case of condensation and it helps mould development increasing allergic risks. Microbial growth, encouraged by high relative humidity, leads to poor indoor air quality (IAQ) and building deterioration. Poor IAQ results in discomfort, health problems and could ultimately lower productivity.

Consequently, the means keeping the indoor RH and CO₂ concentration in a desired levels are necessary and essential to improve building performance in the terms of indoor air quality (IAQ), energy performance and durability of building materials. To achieve acceptable indoor air quality, the American Society of Heating, Refrigerating, and Air-Conditioning Engineers (ASHRAE) recommends two procedures: the Ventilation Rate Procedure, specifying the quality and quantity of ventilation air for a space, and the Indoor Air Quality Procedure, requiring the control of all known contaminants to some specified acceptable levels within a space. Of the two procedures, the Ventilation Rate Procedure is most frequently used (Shaw, 1997). One method that can be used for evaluating the IAQ and hygrothermal comfort is to study the indoor RH and CO₂ levels. As recommend in the European and American standards, the ventilation designed to the occupied zone can be reasonably ascertained by ensuring that the indoor CO₂ concentration is 1000 ppm or less and indoor relative humidity is maintained in the range of 30 to 60 % between winter and summer.

In cold climate conditions, energy requirements for heating of ventilation air are potentially high. A too low ventilation rate results in poor IAQ for the occupants and moisture risk, however, too high ventilation rate results in unnecessary energy consumption. Therefore, this paper aims to couple the both concepts for reducing the energy consumption of an office room made of hemp concrete; more precisely, it focuses on the effect of ventilation strategies on IAQ and energy performance. In this study, the carbon dioxide (CO₂) and indoor relative humidity are considered as indicators for evaluating IAQ and hygrothermal comfort in buildings.

2 PHYSICAL MODELS

Heat and moisture transport in porous building materials

This section presents the model summary for the coupled heat and mass transfer in porous building materials, further details of this one can be found in Tran Le et al (2010). Mechanisms of moisture transport in a single porous building material have been extensively studied. Most of the models have nearly the same origin: the Phillip and de Vries model (1957) and the laws of Fourier, Fick and Darcy. The main difference among them is related to particular assumptions used. Based on Mendes work, we used a model that takes into account liquid and vapour moisture transport (Mendes and Philippi, 2005). Forms of moisture transport depend on the pore structure as well as on the environmental conditions. The liquid phase is transported by capillarity whereas the vapour phase is due to the gradients of partial vapour pressure. With these considerations, the mass conservation equation becomes:

$$\frac{\partial \theta}{\partial t} = \frac{\partial}{\partial x} \left(D_T \frac{\partial T}{\partial x} \right) + \frac{\partial}{\partial x} \left(D_{\theta} \frac{\partial \theta}{\partial x} \right) \quad (1)$$

The phase change occurring within porous materials acts as a heat source or sink, which results in the coupling between moisture and heat transfer. One dimension of the energy conservation equation with coupled temperature and moisture for a porous media is considered and the effect of the adsorption or desorption heat is added. This equation is written as:

$$\rho_0 C_{p_m} \frac{\partial T}{\partial t} = \frac{\partial}{\partial x} \left(\lambda \frac{\partial T}{\partial x} \right) + L_v \rho_l \left(\frac{\partial}{\partial x} \left(D_{T,v} \frac{\partial T}{\partial x} \right) + \frac{\partial}{\partial x} \left(D_{\theta,v} \frac{\partial \theta}{\partial x} \right) \right) \quad (2)$$

where C_{p_m} is the average specific heat which takes into account the dry material specific heat and the contribution of the specific heat of liquid phase and λ is thermal conductivity depending on moisture content.

Heat and moisture transfer in the room air

In order to model heat and mass transfer in the room, we used a nodal method, which considers the room as a perfectly mixed zone characterized by a pressure, a temperature and a

moisture concentration. Nodal method involves equations for air and moisture mass balance, heat balance for ambience and equations describing heat and mass transfer through the walls, additional convection between inside wall surfaces and room ambience and radiation exchange between inside wall surfaces.

Equations describing heat and mass transfer through the walls were presented previously. The energy balance equation for the zone air can be written as:

$$\rho_i c_p V \frac{\partial T}{\partial t} = \Phi_{West} - \Phi_{East} + \Phi_{South} - \Phi_{North} + \Phi_{Bottom} - \Phi_{Top} + \Phi_{Source} \quad (3)$$

This equation describes heat fluxes through the envelope (from walls and openings) and internal heat loads. The moisture level in the room is determined by the moisture transfer from interior surfaces, moisture production rate and the gains or losses due to air infiltration, natural and mechanical ventilation, sources due to habitants of room. This yields to the following mass balance equation for room air:

$$V \frac{\partial \rho_i}{\partial t} = Q_{mWest} - Q_{mEast} + Q_{mSouth} - Q_{mNorth} + Q_{mBottom} - Q_{mTop} + Q_{mSource} \quad (4)$$

Concerning radiation heat exchange between room walls it was computed using the mean radiant temperature method in which the radiative flow between a wall and all the other walls of the room is written as:

$$\Phi_{rad}^{int} = h_r S (T - T_m) \quad (5)$$

Where T_m is the mean radiant temperature of the walls and the value of h_r is expressed by:

$$h_r = 4\epsilon\sigma_0 T_m^3 \quad (6)$$

For the short-wave radiation, we assumed that radiant energy enters the room by pane window and is distributed by the quota of 0,6 for the floor and 0,1 for the other walls.

Model for prediction of indoor carbon dioxide

Consider a well-mixed and mechanically-ventilated space, the mass balance of carbon dioxide concentration can be expressed as:

$$V \frac{\partial C}{\partial t} = Q[C_o(t) - C(t)] + G(t) \quad (7)$$

Where $C_o(t)$ is supply CO₂ concentration (mg/m³); $C(t)$ is indoor carbon dioxide concentration at time t (mg/m³); V is volume space (m³); Q is the volume air flow rate into (and out) of the room (m³/s).

Prediction of percentage perceived discomfort (PD)

The literature shows that indoor humidity affects the acceptability and freshness of indoor air. Investigations showed that complaints about bad indoor air quality increased with a higher enthalpy (H) of the air (Hens, 2011). The enthalpy of moist air is the sum of the enthalpy of the dry air and the enthalpy of the water vapour. The enthalpy of moist air can be described as:

$$H = 1.008T + w(2500.8 + 1.805T) \quad (8)$$

Where: T : dry-bulb temperature of air-vapour mixture (°C); w : humidity ratio, kg of water vapour/kg of dry air.

The statistical relation found between enthalpy and number of dissatisfied (with 20% dissatisfied is handled as the limit) looks like (Hens, 2011):

$$PD = \frac{\exp[-0.18 - 5.28(-0.0333H + 1.662)]}{1 + \exp[-0.18 - 5.28(-0.0333H + 1.662)]} \quad (9)$$

Because at the same time, people generate carbon dioxide and odorous bioeffluents, some authors studied the relationship between the CO₂ level and acceptability of a space in terms of body odor (Fanger et Berg-Munch, 1983). Equation 10 presents the percent of visitor

dissatisfied with the level of body odor in a space in function of indoor CO₂ concentration (CEC, 1992).

$$PD_{Odor} = 395 * \exp[-17.55 * (C_{in} - C_{out})^{-0.25}] \quad (10)$$

The previous equation system has been solved using the finite difference technique with an implicit scheme where details of discretization are explained by Tran Le et al. (2009). Discretized equations were implemented in the environmental simulation SPARK (Simulation Problem Analysis and Research Kernel) allowing to solve efficiently differential equation systems (Sowell et al, 2001; Wurtz et al, 2006). SPARK was developed in collaboration with the Simulation Research Group (SRG) at the Lawrence Berkeley National Laboratory (California, USA).

3 MODEL VALIDATION

Concerning the hygrothermal model, it was fully validated by Tran Le et al. (2011). In this section, the validation for indoor CO₂ concentration prediction model is presented. It is easy to determine the analytical solution from the first order differential equation (7):

$$C(t) = K.e^{-\frac{Q}{V}t} + \frac{QC_o + G}{Q} \quad (11)$$

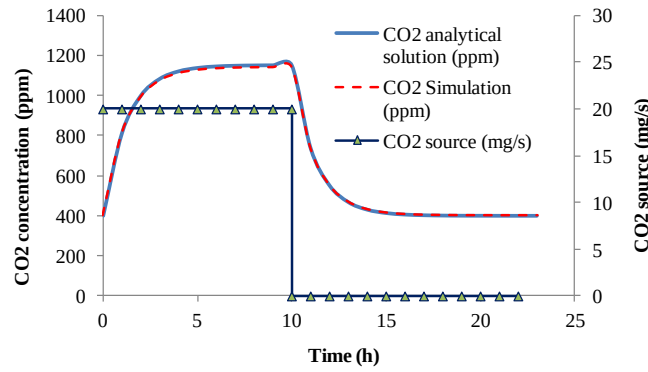


Figure 1: Numerical indoor CO₂ concentration versus analytical solution

To validate the model, we consider a room with a volume of 62 m³ occupied by 2 persons (internal CO₂ source is 20 mg/s) during 10 hours. The ventilation rate and outdoor CO₂ concentration are 0.8 ach and 400 ppm, respectively. As can be seen in Figure 1, a very good agreement between the analytical solution and numerical results is observed showing that the numerical model can be used to investigate IAQ in terms of CO₂ concentration in an occupied space.

4 NUMERICAL STUDY AND DISCUSSION

Description of studied room and simulation conditions

The plan and the section of the studied room are depicted in Figure 2. The room has a space area of 5.5*4 m² and a volume of 61.6 m³. The south facade (a triple-glazed window with U_g=1 W/m²K) is in contact with outdoor conditions while other walls, the floor and the ceiling are considered in contact with heated space at 19°C of temperature. The partitions are multilayered walls made of hemp concrete which are composed of 1.5 cm of lime hemp plaster, 20 cm of hemp concrete and 1.5 cm of lime hemp plaster (from indoor to outdoor).

The hygrothermal properties of hemp concrete and lime hemp plaster obtained from Collet (2004) have been used. The room is equipped with a heat source and a PI controller that keeps operative temperature at 19 °C. Heat convective transfer coefficients are h_{T,e}= 16 W/m².K for the outdoor surfaces and h_{T,i}= 4 W/m².K for indoor surfaces. Mass convective transfer coefficients are calculated from heat convective transfer coefficients by considering the Lewis number equal to 1. The office room is occupied from 8.00 am to 17.00 pm by two persons

(the water vapour, heat and carbon dioxide loads are respectively 140 g/h; 160W and 20 mg/s). The initial CO₂ concentration, temperature and relative humidity (in the walls and the studied room) are respectively 400 ppm, 19 °C and 50 %. The simulations are run for three months in winter using Nancy weather data (in France). The wall is divided into 24 nodes (7 nodes for lime hemp plaster and 10 nodes for the hemp concrete) and the time step is 240 s. In this article, an outside carbon dioxide concentration of 775 mg/m³ (which is equivalent to 400 ppm) is used. A ventilation rate of 0.8 ach is considered as the reference case because this one has been determined based on the ventilation rate required in the office buildings recommended by the European standard (EN 13779-2007).

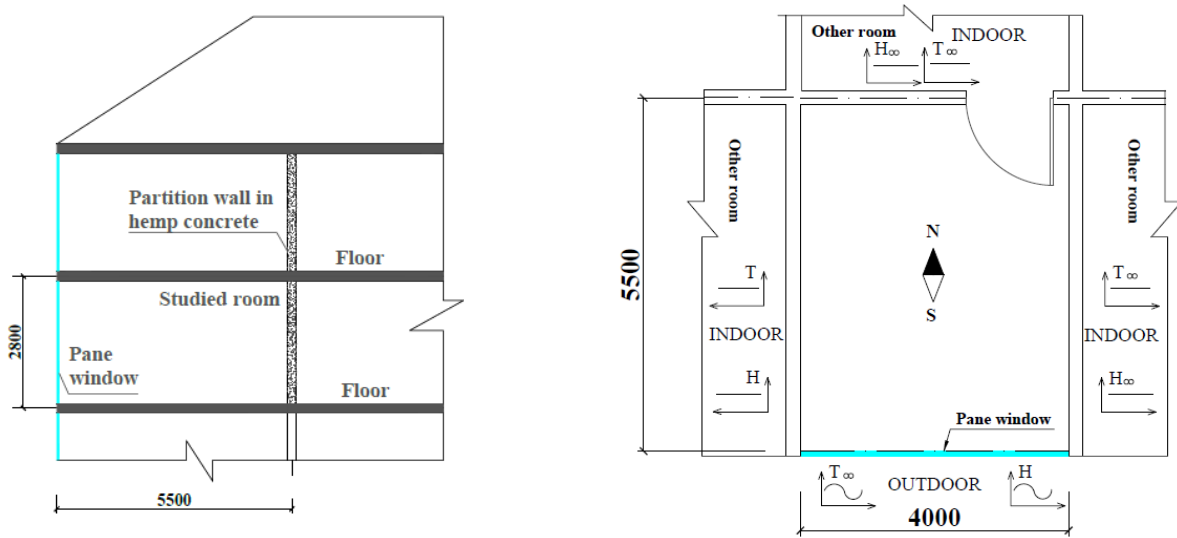


Figure 2: Section and plan of the studied room (unity: mm)

Description of Carbon Dioxide Sensitive Ventilation System (CO₂ SVS) and Relative Humidity Sensitive Ventilation System (RH SVS)

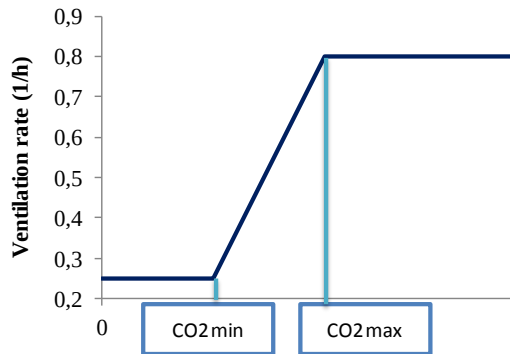


Figure 3: CO₂ sensitive ventilation

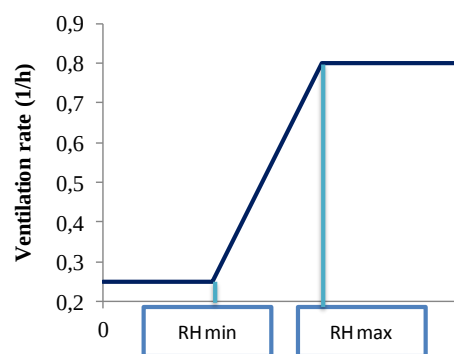


Figure 4: Relative humidity sensitive ventilation

Conventional ventilation introduces a certain amount of outdoor air into the breathing zone to dilute the concentration of contaminants for an acceptable indoor air quality (IAQ). A fixed ventilation rate which is used over time results in wasted energy when the space is unoccupied. Demand controlled ventilation (DCV) had been developed to consider both IAQ and energy efficiency by providing outdoor air to a space based on actual human demand (Owen et al, 2010). In this article, two ventilation strategies which match the outdoor airflow rate of a ventilation system to real-time occupancy as indicated by indoor CO₂ concentration (for CO₂ sensitive ventilation system: CO₂ SVS) or indoor relative humidity (RH sensitive

ventilation system: RH SVS) are studied. It is noticed that the final one is now widely used in France. The description of these ventilation systems is shown in Figure 3 and Figure 4:

- ✓ During the hours of normal occupancy, the ventilation rate must be adequate to maintain the indoor CO₂ concentration or relative humidity level lower than or equal to a maximum target value (called CO₂max or RHmax).
- ✓ During the unoccupied period, the ventilation rate is maintained at 0.25 ach when the indoor CO₂ concentration or relative humidity is lower than 500 ppm or 30 %, respectively.

Effect of carbon dioxide sensitive ventilation system (CO₂ SVS) and relative humidity sensitive ventilation system (RH SVS) on indoor CO₂ concentration and PD

Figure 5 shows the indoor CO₂ concentrations for different ventilation strategies. The higher target value is, the higher indoor CO₂ concentration is. It can be seen that indoor CO₂ concentration of two cases Target_1000 ppm and reference case (ventilation rate is constant and equal to 0.8 ach) is very close. When the target value CO₂max passes from 1000 ppm to 1200 ppm, the maximum indoor CO₂ concentration increases from 1140 ppm to 1280 ppm. Regarding the RH SVS, when RHmax changes from 60 % to 70 % the maximum indoor CO₂ concentration changes from 1630 ppm to 1730 ppm.

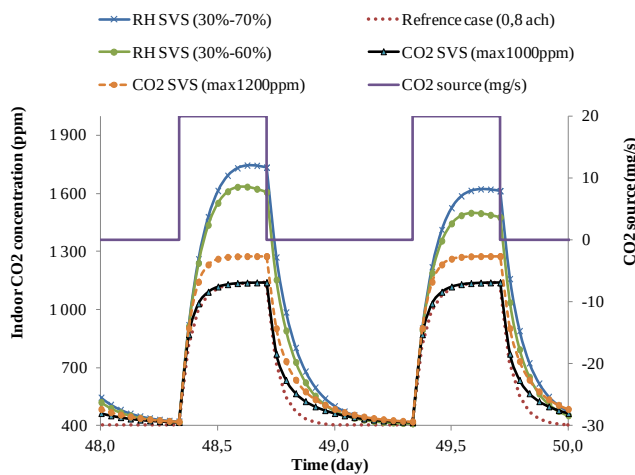


Figure 5: Indoor CO₂ concentration as function of ventilation strategies

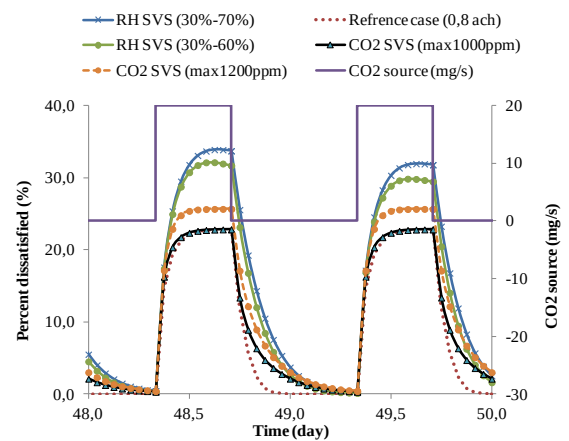


Figure 6: Percent dissatisfied (body odor) as function of ventilation strategies

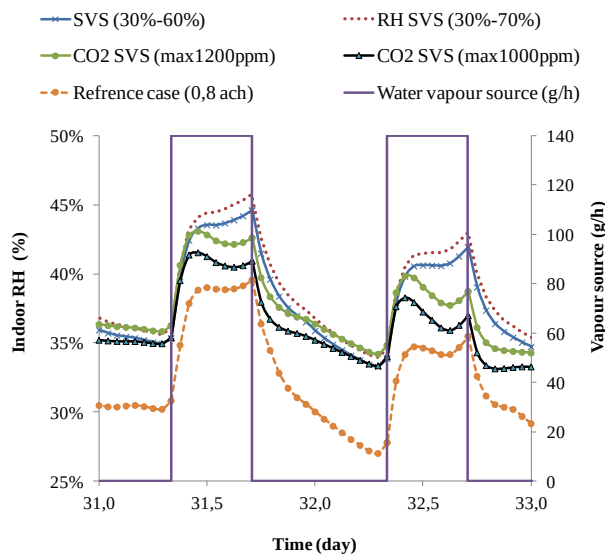


Figure 7: Indoor RH as function of ventilation strategies

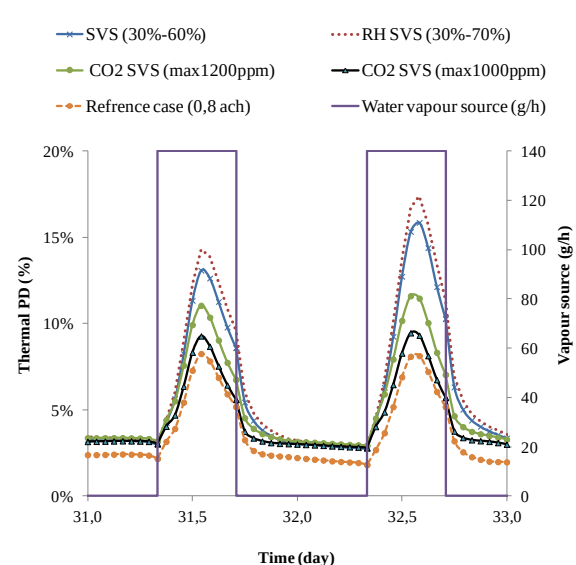


Figure 8: Thermal percent dissatisfied as function of ventilation strategies

Compared to the RH SVS, the CO₂ SVS results in a lower indoor CO₂ concentration. Numerically, the maximum indoor CO₂ concentration is 1140 ppm for CO₂ SVS (max 1000 ppm) and 1630 ppm for RH SVS (30%-60%) so a difference of 30% was observed. Note that indoor CO₂ concentration affects the acceptability of a space in terms of body odor which can be expressed via the percent dissatisfied index (see Figure 6). The result showed that the use of CO₂ SVS can reduce the percent dissatisfied in terms of body odor from 34% to 23 % in comparison with RH SVS. Figure 7 and Figure 8 illustrate the effect of ventilation strategies on indoor relative humidity and the thermal percent dissatisfied. It can be seen that, in comparison with the ventilation system with constant air flow rate over time, demand control ventilation systems result in higher indoor RH and thermal PD index during the occupied period. The indoor RH is always in the comfortable zone and the thermal PD is lower than 20%, except the reference one (0.8 ach) where indoor RH is sometimes smaller than 30% at the beginning of occupied time (see Figure 7).

Effect of carbon dioxide sensitive ventilation system (CO₂ SVS) and relative humidity sensitive ventilation system (RH SVS) on ventilation rate and on energy consumption

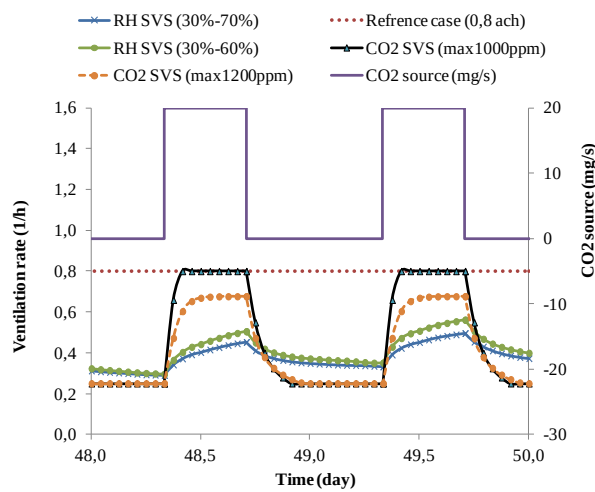


Figure 9: Ventilation rates for different ventilation strategies

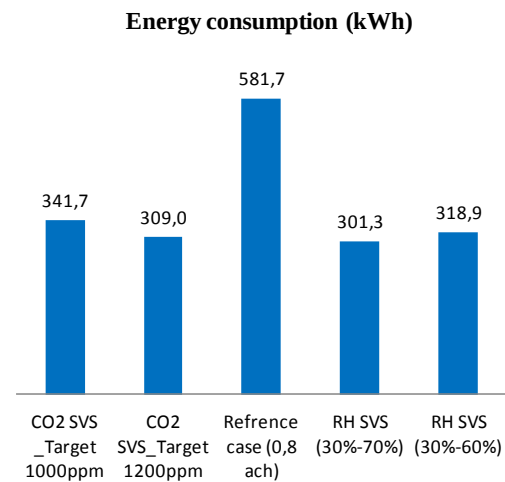


Figure 10: Effect of different ventilation strategies on energy consumption

The ventilation rates for different ventilation strategies are shown in Figure 9. As can be seen from this figure, the one for the CO₂ SVS is higher than for the RH SVS during the occupied and but it is smaller during the unoccupied period.

Figure 10 illustrates the effect of ventilation strategies on energy consumption. It is shown that decreased ventilation rate can reduce the energy consumption. In addition, the use of the RH SVS results in smaller energy consumption compared to the CO₂ SVS (target 1000 ppm). For example, during the three months investigated, energy consumption for the RH SVS (30%-60%) and the CO₂ SVS (target 1000 ppm) is respectively 318.9 kWh and 341.7 kWh, with a difference of 6.7%. Furthermore, in comparison with the system using fixed ventilation rate (reference case 0.8 ach), the RH SVS (30%-70%) can reduce energy consumption about 48 % without degrading indoor air quality.

5 CONCLUSIONS

This article focuses on the effect of carbon dioxide and relative humidity sensitive ventilation systems on indoor air quality, hygrothermal and energy performance in an office room made of hemp concrete. A coupled model for predicting hygrothermal performance and indoor air quality in a room is presented and implemented in the environment SPARK which is suitable for complex problems using finite difference technique with an implicit scheme. After being validated, the models are used to analyze the building performance in terms of IAQ, energy

consumption, hygrothermal comfort in a room occupied by two persons during three months in winter season using Nancy weather data (in France).

The results showed that an air flow rate as required in European standard can ensure a good indoor air quality during the occupied period. It is shown that the relative humidity sensitive ventilation system which is now widely used in France results in higher percent dissatisfied in terms of body odor and thermal comfort in comparison with the CO₂ sensitive ventilation system because moisture buffering capacity of building materials can reduce indoor RH variations.

This study confirmed that the use of demand control ventilation system is a very efficient way to reduce the energy consumption. For our studied cases, compared to the case using the fixed ventilation rate, demand control ventilation system can reduce the energy consumption about 48 % without degrading indoor air quality.

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